

Loose Puzzle Pieces

Kaluza-Klein and other established building blocks of the time-mass duality

Johann Pascher
ORCID 0009-0000-6518-4064

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Loose Puzzle Pieces: Kaluza-Klein and Other Established Building Blocks of the Time-Mass Duality

Abstract

Anyone who sees $\tilde{T} \cdot m = 1$ for the first time often recognises something familiar in it, most often Kaluza-Klein. That is not an objection but a finding: almost every building block of FFGFT has been in the literature for decades, scattered, however, across relativity, quantum mechanics, group theory, string theory, number theory and phenomenology. This document places eleven such building blocks: Kaluza-Klein, de Broglie's internal clock and the Compton time, the Zitterbewegung, Pauli's objection to a time operator, the relational approaches to time, Wigner's mass as a Casimir invariant, the Casimir value of $SU(3)$, Heaviside-Lorentz units, orbifold compactification, finite fields with their Frobenius automorphism, and the Koide formula. For each building block it records what it already contained, which piece was missing, and where in the corpus it fits in. The order matters: these building blocks were not the foundation of FFGFT. The theory arose independently of them from the relation $\tilde{T} \cdot m = 1$; the building blocks were found in the literature only afterwards and fit into a picture that was drawn without them. The result is a map of the places at which the theory could already have been found.

Status markers

[SETZUNG]	Motivated input, not a derivation.
[B]	Algebraically proved (verification script, exact arithmetic).
[K]	Numerically confirmed.
[S]	Structurally plausible, identification open.
[E]	Established mathematics or physics, adopted.
[Q]	Taken from the literature, with source.
[X]	Checked and negative.

The markers in this document give the status *in the corpus*; the document itself derives nothing new but places things in context. Verification script: `pruef_390_puzzleteile.py`.

1 The question

In the introductory conversation on the website, the sceptical physicist asks right at the start whether this does not sound like Kaluza-Klein. The answer is yes, and the question can be generalised: if the duality of mass and time was already known, why was it not thought through to the end?

For Einstein, the corpus has already answered this question. Doc. 312, chapter “Why Einstein did not see $\tilde{T} \cdot m = 1$ ”, shows that the ingredients were on his desk and that the missing step was not a formula but the change of status of a known formula into a principle. The present document does not repeat that account. It widens the view: not just one ingredient but almost all of them were available, each in a different field, each treated as a tool or a curiosity. The following sections go through them in three groups: the rolled-up direction, mass as structure, and the geometry with its algebra.

On the order. None of the following building blocks was a starting point of FFGFT. The theory was developed from $\tilde{T} \cdot m = 1$ and the geometry of the carrier; the literature references were looked for and found only afterwards. Where it says below that a building block fits in at some place in the corpus, this is to be read as agreement in hindsight, not as origin. This is precisely what gives the map its value: pieces that arose independently of one another and independently of FFGFT fit into the same picture.

2 Time and mass: the rolled-up direction

Kaluza-Klein (1921, 1926)

What it contained. Kaluza extended general relativity by a fifth dimension, and Klein rolled it up into a circle of radius R **[Q]** [1, 2]. On a circle only integer windings are allowed; from this a mode tower with $m_n \propto n/R$ arises, and momentum along the circle direction appears in four dimensions as mass. Klein thereby also obtained the quantisation of charge. So by 1926 the core statement was in place that a compact direction generates mass as the inverse of its period.

What was missing. The rolled-up direction was spatial and was added to spacetime. The radius R remained a free parameter, and the predicted tower of heavy excitations was never found. As Doc. 311 puts it, one had gained a scale and bought a parameter along with it.

Where it fits in the corpus. In the corpus, $\tilde{T} \cdot m = 1$ is explicitly the Kaluza-Klein relation of the fourth torus direction, the mass dimension s_m^1 (Doc. 330, Doc. 270). The direction is

timelike: what holds there for a rolled-up spatial direction holds here for rolled-up time, with mass in the role of the mode index (Doc. 307; likewise Doc. 231). The radius is not a free modulus, because $\tilde{T} \cdot m = 1$ binds the direction; Doc. 311 summarises this as “scale without modulus”, and Doc. 318 sets the three routes side by side. There is no additional tower to wait for: the discrete spectrum of the compact direction already is the observed particle spectrum (Doc. 307). Locally, Lorentz invariance holds without restriction; the compactness shows only globally (Doc. 312).

De Broglie’s internal clock and the Compton time (1924, 2013)

What it contained. De Broglie assigned to every particle an internal clock with frequency $\nu = mc^2/h$ [Q] [3]. In natural units this is $T \cdot m = 1$ in raw form. The Compton time $\hbar/(mc^2)$ has been a standard notion ever since, and in 2013 Lan, Müller and co-workers built a clock whose rate follows directly from the mass of an atom [Q] [4].

What was missing. The clock became the wavelength; the clock itself dropped out as a derived scale. The history of this step is told in Doc. 312 and is not repeated here.

Where it fits in the corpus. The de Broglie relation appears on the mass direction as $\lambda_4 \cdot m = 2\pi$, exact mode by mode (Doc. 314); the clock character of every mass is the starting point of the theory, not its consequence (Doc. 365).

The Zitterbewegung (1930, 1990)

What it contained. Schrödinger found in the Dirac equation a rapid oscillation of the electron’s position with angular frequency $2mc^2/\hbar$ [Q] [5]. Hestenes interpreted it as an internal circular motion and explicitly called the electron a clock [Q] [6].

What was missing. The Zitterbewegung remained a peculiarity of single-particle Dirac theory, regarded in the field picture as interference of positive- and negative-energy parts; no geometric direction in which it takes place was assigned to it.

Placement. The corpus has not drawn on the Zitterbewegung so far. It belongs in this series because it shows the same reciprocity of mass and rate in another language. An identification with the circle direction S_m^1 is not claimed here; the factor 2 in the frequency and the role of the negative-energy parts would first have to be clarified [S].

Pauli’s objection to a time operator (1933)

What it contained. Pauli showed that no self-adjoint time operator can exist for a Hamiltonian that is bounded from below [Q] [7]. For decades this was taken as a reason to keep time out of the state space.

What was missing. The objection presupposes open time. Compact time behaves like an angle variable; its conjugate spectrum is discrete, as for angular momentum conjugate to an angle. This gap in the argument was not exploited consistently.

Where it fits in the corpus. Doc. 307 and Doc. 334 record that Pauli’s objection does not apply to the rolled-up view: for compact time the Pauli condition is not satisfied, and the discreteness of the modes becomes content rather than obstacle.

Relational and timeless approaches (1983 onwards)

What it contained. Page and Wootters obtain time from correlations in a static overall state [Q] [8]; the Wheeler-DeWitt equation and Rovelli’s relational time proceed similarly. Kaluza-Klein variants with compact time and Bars’ two-time physics have been studied [Q] [9].

What was missing. The mainstream takes time *out* as something fundamental. The opposite route, taking time *in* as a compact dimension and obtaining the mass modes from it, has hardly been pursued.

Where it fits in the corpus. Doc. 307 draws exactly this line: shared is the rejection of an external parameter time; different is the positive proposal, compact geometric time instead of relational or thermal time.

3 Mass as structure

Wigner: mass as a Casimir invariant (1939)

What it contained. Wigner classified particles as irreducible representations of the Poincaré group; mass is a Casimir label there, hence already a structural eigenvalue and not an attached number [E] [10].

What was missing. The classification does not say which values this label takes; the mass values remain inputs.

Where it fits in the corpus. Doc. 307 cites Wigner as the reference showing that mass as a structural quantity is well familiar; Doc. 339 uses the Wigner classification as a standard step in separating the massive and massless sectors. In FFGFT the mass values themselves come from the compact direction, not from the Poincaré group.

The Casimir value of $SU(3)$

What it contained. The quadratic Casimir operator of the fundamental representation of $SU(N)$ has the value $(N^2 - 1)/(2N)$, for $SU(3)$ therefore $4/3$ [E] [11]. The value is textbook material of colour dynamics.

What was missing. Nobody had a reason to connect this value with a mode count of a compact geometry.

Where it fits in the corpus. In Doc. 324, $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4 = 4/30000$; the formula is [B] there, the Fourier mode count [K]. The same value follows in Doc. 338 from group orders of finite fields (section 4).

Heaviside-Lorentz units (1893)

What it contained. Heaviside introduced rationalised units in which the factors 4π disappear from the field equations [Q] [12]. With $\hbar = c = \varepsilon_0 = 1$, $\alpha = 1$ becomes a choice of units.

What was missing. The choice of units remained a computational convenience; that $\hbar$ and c are mere conversion factors became self-evident only with the practice of natural units and the SI reform of 2019 (Doc. 312).

Where it fits in the corpus. Doc. 365 sets up the theory in this basic form: $\alpha = 1$ is not a setting but a choice of units, and the electromagnetic quantities become derived forms of mass and time. The familiar number 137 arises only in the transition to SI units; its agreement with the measured value depends on the anchor E_0 (Doc. 338).

4 Geometry and algebra

Orbifold compactification (1985)

What it contained. Dixon, Harvey, Vafa and Witten compactified string theory on orbifolds, tori divided by a finite group; $T^6/\mathbb{Z}_3$ was one of the first examples [Q] [13]. The fixed points

of the group carry sectors of their own there, and the number of generations becomes a topological quantity.

What was missing. Six additional spatial dimensions and a vast number of possible compactifications; time stays outside. Doc. 311 applies the corpus's two conditions to string theory: the compact directions are exemplarily compact, all information is rebooked into conserved quantum numbers, and what remains open is what the directions carry.

Where it fits in the corpus. The carrier is $T^4/\mathbb{Z}_3$ on the D_4 lattice, four directions instead of six additional ones, with the time-mass direction among them (Doc. 330, Doc. 314). The $\mathbb{Z}_3$ acts freely on all non-zero momenta; in position space the orbifold has nine fixed points (R131 in Doc. 190). Torus, $\mathbb{Z}_3$ and D_4 are settings **[SETZUNG]** (Doc. 365).

Finite fields and the Frobenius automorphism

What it contained. The finite fields $GF(p^n)$, their cyclic multiplicative groups and the Frobenius automorphism $x \mapsto x^p$ have been fully described since Galois **[E]** [14]. They were not connected with elementary particle physics.

What was missing. A physical structure in which the prime 3 and its extensions occur naturally.

Where it fits in the corpus. The $\mathbb{Z}_3$ of the orbifold supplies this structure. The Frobenius automorphism of $GF(9)$ corresponds to the sector pairing $k \leftrightarrow -k$ **[B]** (Doc. 336). Charge quantisation follows from the Legendre symbol modulo 13 **[B]** (Doc. 346), the three generations arrange themselves in the Frobenius orbits of $GF(27)^*$ **[B]** (Doc. 348), and the identity $(r_\mu/r_e)^2/\xi = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 43200$ connects the lepton ladder with the group orders **[K]** (Doc. 338). The hypercharge of the charged leptons is still open here (R146 in Doc. 190).

The Koide formula (1981–1983)

What it contained. Koide found that the three charged lepton masses satisfy the relation $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ to about 10^{-5} **[Q]** [16]. For more than forty years the formula remained unexplained, a curiosity of phenomenology.

What was missing. A reason for the value $2/3$ and for the phase that the formula requires in its cosine form.

Where it fits in the corpus. The Koide phase is $\theta = 2/9$ **[B]**, the same quotient as the squared modulus of an icosahedral projection (Doc. 367). The amplitude $d/c = \sqrt{2}$ has the right value but is not yet derived from the geometry and is treated as a setting **[S]** (Docs. 353, 367; R150 in Doc. 190). A further puzzle piece fits in here afterwards: Foot showed in 1994 that $Q = 2/3$ means an angle of 45° between the vector of the $\sqrt{m_k}$ and the axis $(1, 1, 1)$ **[Q]** [15]. In the language of the corpus this is the equipartition of the mass sum between the $\mathbb{Z}_3$ -symmetric mode and the pair of non-trivial modes **[B]** (Doc. 353). The open question is thus no longer why $\sqrt{2}$ in particular, but why the mass structure divides half and half between these two parts. The exact pair $(\sqrt{2}, 2/9)$ is excluded for m_μ/m_e (Doc. 369). The formula is thus placed, but not completely closed.

5 Why the pieces did not come together

Three reasons run through all the sections.

The pieces lay in different fields. Kaluza-Klein belongs to gravitation and later to string theory, the internal clock and the Zitterbewegung to quantum mechanics, Wigner and the Casimir value to group theory, finite fields to number theory, the Koide formula to

phenomenology. Whoever works in one of these fields rarely has reason to read the others as parts of the same picture.

Each piece was classified as a tool or a curiosity. The Compton time counted as a conversion, Heaviside's units as a convenience, the Zitterbewegung as an artefact, the Koide formula as a coincidence. The missing step was the same each time as with Einstein: to take a known relation not as a derived quantity but as a principle (Doc. 312).

Two switches were set wrongly. Pauli's objection kept time out of the state space, and Kaluza-Klein rolled up a spatial instead of the temporal direction. Both switches could have been reset with the same insight: compact time escapes the objection, and it delivers the mass modes without a free radius.

There is also a methodological point. That the pieces fit into the picture is a structural agreement, not a numerical one. The more complex the quantum numbers, the closer one comes to the noise floor, where a matching combination can always be found; no status is assigned there (Doc. 342, Doc. 343). Scheme-dependent quantities of the Standard Model such as Λ_{QCD} are excluded from any claim of derivation. The puzzle pieces therefore hold only where they concern simple, convention-free ratios, and that is exactly where the places in this document lie.

6 Overview

Building block	Year	What was missing	FFGFT in the corpus
Kaluza-Klein	1921/26	spatial direction, free radius, unobserved tower	Doc. 330, 270, 307, 311, 318
De Broglie, Compton time	1924/2013	clock only as a derived scale	Doc. 312, 314, 365
Zitterbewegung	1930/1990	no geometric direction	not used so far [S]
Pauli's objection	1933	presupposes open time	Doc. 307, 334
Relational time, two-time physics	1983 on	time taken out instead of in	Doc. 307
Wigner	1939	mass values remain inputs	Doc. 307, 339
Casimir value $SU(3)$	textbook	no link to modes	Doc. 324, 338
Heaviside-Lorentz	1893	only a computational convenience	Doc. 365, 312
Orbifold	1985	six spatial directions, time outside	Doc. 330, 314, 311; R131
Finite fields, Frobenius	Galois	no physical structure	Doc. 336, 338, 346, 348
Koide, Foot	1981–83, 1994	no reason for $2/3$ and the phase	Doc. 367, 353, 369; R150

7 Conclusion

FFGFT connects to established physics and mathematics at eleven places, and at each of them it could have been found. Kaluza-Klein had the compact direction, de Broglie the clock, Pauli the objection whose gap is compact time, Wigner mass as structure, group theory the value $4/3$, string theory the orbifold, number theory the finite fields, and Koide the puzzle they help to solve together. FFGFT is not built from these pieces. It arose independently from time as a compact direction carrying the mass modes, and from the algebra that

follows from it with the one parameter ξ . The building blocks were found in the literature only afterwards, and they fit in. That does not speak against the theory but for it: pieces that arose independently of one another fit into a picture that was drawn without them.

Verification script: `2/python/Dok390_Skripte/pruef_390_puzzleteile.py` — 59/59.

Tool and method

Claude (Anthropic) was used as a tool for: compiling the references in the corpus, writing and running the verification script, and working out the $\text{\LaTeX}$ sources according to the author's specifications. The verification script backs every corpus reference in this document with a text check in the named chapter file; if a check fails, the reference is not used.

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